

What's the definition of IoT?

- A cyber-physical system consisting of a large number of individually and globally addressable devices which have a significant portion of their logic running elsewhere

What are often some of the consequences with IoT?

- Siloed access
- Lots of external dependencies
- Lack of security

What is the definition of big data?

- A class of processing regimes which primary property is that it relies on a body of data which expands at a rate where processing has become nontrivial

In cloud,fog,edge and mist, what is correct?

- When going from edge to cloud, processing power increases
- When going from cloud to edge, latency decreases
- When going from cloud to edge, privacy improves

What languages are typical classified as high-level languages?

- DSL
- Elixir
- Erlang

What languages are typical classified as low-level languages?

- Assembly
- Zig
- C
- Rust

What are typical requirements for programming languages used in IoT, in terms of cloud and fog?

- Handle high amount of concurrency
- Have a high degree of availability
- Offer robust low response times
- Be mature
- Access to frameworks for modeling information (graphs, ontologies)

What are popular BEAM-based languages for IoT, in Cloud and Fog

- Elixir
- Erlang

Back in the 80's, what was some of the properties needed for new platforms?

- Distribution
- Soft real-time properties
- Close to zero downtime

What are some of the imperative programming traits?

- Focus on data
- Code is an recipe, data is the ingredients
- You tell the steps and how it should be executed
- The focus is how the result is reached

What are some of the declarative programming traits?

- The focus is on what the program shall achieve
- Focus on result
- Control flow is not explicitly described
- You define what the result is

What form is functional programming?

- Declarative programming

What are some properties of functional programming?

- Data are immutable
- Everything is an expression
- Functions are values
- Functions are composable through their arguments
- Function calls are cheap
- There are no loop (in principle)
- Functions are first class citizens

When is a functional language considered pure?

- Only supports pure functions
- No side-effects
- No randomness

What are pure functions?

- Functions whose results are depends only on the value of their parameters
- When they are idempotent

What are some properties of pure functions?

- Two functions that are pure, cannot affect each other

- Functions can be executed in parallel
- Easy to compile
- Easy to test
- Hard to write

What is one of the key elements highlighted about Elixir?

- Pattern matching
- Pipelines
- Anonymous functions
- Higher-Order functions

What is the convention of functions with naming "function"?

- Returns a tuple, with the first element being an atom indicating success or failure

What is the convention of functions with naming "function!"?

- Returns the result directly, or raises an exception if it fails

What is the convention of functions with naming "function?"?

- Returns a boolean indicating success or failure
- Works as a "test"

What is an actor in the context of Elixir?

- A process
- Consists of state
- Consists of Functionality
- Consists of a message queue

What are some properties of actors in the actor model?

- Actors are isolated from each other
- Actors run concurrently
- Actors communicate through message passing
- Actors are inactive as long as their inbox is empty
- Messages are consumed FIFO-style and processed strictly sequentially
- Actor is an "Unit of concurrency"

In elixir and actor model, what is synchronous communication?

- A call
- Waits for other process to respond

In elixir and actor model, what is asynchronous communication?

- A cast
- Does not wait for other process to respond

What is a supervisor in the context of Elixir?

- A type of process
- responsible for administrating the lifecycle of a number processes
- Starts and stops processes
- Handle crashes of processes

What type of restart strategies are available in for the supervisor in Elixir?

- One for one
- One for all
- Rest for one

In regards to restart strategies, what does "one for one" mean?

- Only the crashed process is restarted
- Other processes are not affected

In regards to restart strategies, what does "one for all" mean?

- If one child process crashes, all other child processes are restarted

In regards to restart strategies, what does "rest for one" mean?

- The process that crashed is restarted
- All process listed after the crashed process are restarted

Where is the state of processes stored in Elixir?

- In a sandboxed process

What is defensive programming and its traits?

- Handling errors gracefully
- Handling errors using if statements
- Handling errors using exception handling

What is offensive programming and its traits?

- Let it fail
- It's okay for your code to fail

In terms of cyber-physical systems, what is the approach "A narrow waist"?

- Having a common interface and interface model between applications and buildings

What is a timeseries database?

- Persists a values, which are indexed by timestamps

- Typically stored as a B+ tree
- Used for fast range lookups

What is an ontology?

- Formal naming and definition of types, properties and interrelationships of entities
- Exists in a particular domain

What is correct about ontologies?

- Formalized language
- Things are defined through the type, parameters and relationships to other things
- Limited coverage
- Focus on a particular field
- Models can span multiple ontologies
- The ontology in a model

What are some of the challenges when defining ontologies

- Envisioning how it will be interpreted
- How writing to the model affects model consistency
- Think in terms of lower abstractions

What do you have to think of, when producing the representation model of an ontology?

- Envisioning how it will be interpreted

What do you have to think of, when producing the data, adhering to the data model of an ontology?

- How writing to the model affects model consistency

What do you have to think of, querying a model of an ontology?

- Think in terms of lower abstractions

What does RDF stand for?

- The Resource Description Framework

What is RDF?

- An ontology defined through triples
- Defined through Subject, predicate & object
- Namespaces
- Subclasses and subproperties
- The information lies in the triple

Why does RDF use namespaces for entities?

- To avoid naming collisions

How are RDF namespaces done?

- Realized through compact URI (CURIE)
- Through prefixes

To work with RDF, what query language is used?

- SparQL

In RDFS and OWL (Ontologies), how are relationships defined in the DAG of types?

- Through their use
- Through their place

In SparQL, what happens if you drop RDF support, when you support RDF and OWL?

- Removes the ability for any entity to take any role in a triple
- Allows modeling the data as a graph

When working with ontologies and RDF, what does Closed world assumption mean?

- The presumption that any statement that is true is also known to be true
- Require all facts to be known
- It is possible to validate that all properties hold

When working with ontologies and RDF, what does Open world assumption mean?

- The assumption that the truth value of a statement may be true irrespective of whether or not it is known to be true
- Accept that you don't know all facts
- A property that needs to be satisfied could reside outside the set of known facts
- Impossible to verify that properties hold
- Possible to check that no restrictions are violated

When looking at ontology schemas, what classifies RDF?

- Concept
- Model

When looking at ontology schemas, what classifies RDFS?

- Open World
- Classification

When looking at ontology schemas, what classifies SHACL?

- Closed World
- Validation

What is a validator in RDF ontologies?

- Tests the stated rules of the applied ontologies

What can happen, if you don't have a validator in RDF ontologies?

- Spelling mistakes are accepted
- Entities can be connected in any way
- Entities can be floating

What is a reasoner in RDF ontologies?

- An element that applies rules

When can you apply a reasoner in RDF ontologies?

- When inserting
- On triple store before querying
- While querying

When is a graph a weighted graph?

- When all edges have a weight

When is a graph query language a real graph query language?

- The level of abstraction refers to nodes, edges, properties and labels
- Paths can be matched
- A suite of graph algorithms can be referenced

In Neo4J, what is the data model of nodes?

- A set of labels
- A key-value store containing named properties
- A map from relationship type to set of outgoing relationships
- A map from relationship type to set of incoming relationships

In Neo4J, what is the data model of edges (relationships)?

- A single type
- A key-value store containing named properties
- A source node
- A destination node

What are some of the reasons to use C in IoT?

- Garbage collection is problematic in embedded systems
- Fast
- Low memory footprint
- Easy to access low-level operations
- Include important high-level abstractions
- Simple language

What are the build phases in C?

- Pre-compilation
- Compilation
- Linking
- Execution

In C, what is the issue if you just make an array, without using systems, such as malloc?

- The array may contain garbage values

In C, what is a pointer?

- Variable contains only an address
- Contains an integer value
- Can be obtained by using the & operator
- Can be dereferenced by using the * operator

What is the definition of Voltage?

- The difference in electric potential
- Measured between two points
- Measured in Volts (V)

What is the definition of transducer?

- A device which converts energy from one form to another

What is the definition of serial communication?

- A way of transferring a chunk of data one bit after the other

In RS232, what are the different wires used for?

- Communication in one direction
- Communication in the other direction
- Ground

In RS232, what are the elements in the frame format?

- 1 start bit
- A character spanning 7-8 bits

- 1 parity bit
- Stop bits

When choosing machine-readable formats in serial communication, what is correct?

- Makes it simple to parse

When choosing human-readable formats in serial communication, what is correct?

- Makes it simple to debug

What are some of the options for dispatch mechanisms in serial communication?

- Lookup using table
- Lookup using map
- Chain of branches
- Deterministic finite automaton

What is the definition of sampling interval or sampling period (T)?

- The time between two samples

What is the definition of sampling frequency or sampling rate (f_s)?

- The number of samples obtained within 1 second

How frequent do you have to sample to be guaranteed to be able to reconstruct a signal?

- Twice as fast as the fastest frequency contained in the original signal

What is the theorem called, which requires double the sampling frequency to reconstruct a signal?

- Nyquist Theorem

What are possible resampling methods?

- Downsampling
- Upsampling

What are some errors that can occur in resampling?

- Aliasing
- Cutting off extreme values
- Remove fluctuations

In analog to digital conversion, what are some of the steps?

- Flush
- Wait
- Sample
- Hold
- Compare

In analog to digital converters, what are some of the sources of noise?

- The sensor itself
- Electromagnetical signals
- AD conversion

What is the noise floor?

- The inherent noise in a given situation
- The minimum signal that can be detected

What is signal to noise ratio (SNR)?

- A measurement that tells us how large a fraction of a signal consists of noise

What happens if you average two samples?

- The signal is doubled
- Many types of noise will cancel out
- Signal to noise ratio (SNR) goes up
- Signal to noise ratio (SNR) is improved

What is an outlier?

- An observation which lies 'far' away from the brunt of distribution

What can happen, if you remove important outliers?

- Data about an unexpected event is lost
- Information about something thought impossible is lost
- A black swan event is lost

Given a 12 bit ADC, with a 3 bit noise floor, how many bits should be used for the signal?

- 9 bits

Given a 12 bit ADC, with a 3 bit noise floor, sampling at 10 kHz, what is the theoretical required bandwidth?

- 90 kbps

How are radio waves classified?

- Frequency
- Wavelength

When working with wireless communication, what is true?

- Lower frequency gives longer range
- Higher frequency gives shorter range
- There exists an open and closed spectrum
- The signal extends according to the inverse square law
- Media is shared
- It is not possible to avoid collisions

What are some of the consequences of the inverse square law?

- The intensity of a signal drops polynomially with distance
- It is infeasible to send a signal over a long distance

What happens when there is not enough signal strength to noise ratio (SNR)?

- Bit errors start to occur

What is the definition of a collision in wireless communication?

- Happens when two or more signals overlap in both time and space
- High probability of one, some or all signals are lost at the place of collision

What are possible network topologies in wireless communication?

- Line
- Star
- Ring
- Tiered
- Tree
- Mesh

What is true about packets in wireless communication?

- Packets are usually made up of individual frames
- Frames are often dropped due to detected corruption
- Frames are often dropped due to complete lack of reception
- ACK can be used to acknowledge that a frame has been received

In the OSI model, what is the Physical layer do?

Also known as Layer 1

- Electrical link

- Audio link
- Radio link

In the OSI model, what is the Data link layer do?

Also known as Layer 2

- Reliable delivery of frame to local named receiver

In the OSI model, what is the Network layer do?

Also known as Layer 3

- Multi-node routing

In the OSI model, what is the Transport layer do?

Also known as Layer 4

- Connection-orientation
- Multiplexing
- Flow control

In the OSI model, what is the Session layer do?

Also known as Layer 5

- Session management of connections

In the OSI model, what is the Presentation layer do?

Also known as Layer 6

- Convenient translation
- Encryption
- Coding

In the OSI model, what is the Application layer do?

Also known as Layer 7

- API

In the OSI model (expanded), what is the individual layer?

Also known as Layer 8

- Application user

In the OSI model (expanded), what is the organization?

Also known as Layer 9

- Organizational framework

In the OSI model (expanded), what is the government layer?

Also known as Layer 10

- Rules for organizations and compliance

What are some key properties of a typical IoT cloud?

- Large number of incoming data streams
- Large number of customers
- Similar processing across multiple customers
- Expectation of privacy
- Small number of processing products presented to the customer
- Possibly large number of outgoing data streams

What is a stream process?

- An independent piece of logic
- Associated with a state
- Associated with zero or more input streams
- Associated with zero or more output streams

What is true about stream processing?

- A node consumes inputs in order to produce outputs
- A source, is a node only producing output streams
- A sink, is a node only consuming input streams

What are some specific properties of stream processing, that you should think of, when designing?

- Flow control
- Consumability
- Trigger condition
- Trigger arbitrariness
- Batch size
- Fault tolerance

What are some concerns that can arise in stream processing?

- Context switching
- Side effects
- Networking

What are some of the steps, that data has to go through, when being ingested?

- Categorization

- Validation
- Sanity check

What is a typical issue with message queues?

- If one element is down, it may break the entire pipeline

What is true about the publish-subscribe pattern?

- A communication pattern
- Messages can be published through some name on a broker
- Clients can subscribe to names on a broker
- Whenever a message is published, all clients subscribed to that name are notified
- Contains a publisher, broker and client
- There exists dumb brokers and smart brokers
- There exists dumb clients and smart clients

In regards to dumb and smart brokers and clients, what is true?

- Kafka is close to dumb broker, smart client
- MQTT is in between a dumb broker and a smart broker, and a dumb client and a smart client
- "Data Portal" is close to smart broker, dumb client

In regards to publish-subscribe and message queues, what is true?

- Kafka subscribe to exact topics
- MQTT subscribe using query over topic pattern
- sMAP subscribe using query over per-stream key-value store

Why is Kafka not often used in IoT?

- Due to its large client footprint
- Due to its complexity
- Due to complexity in topic discovery

What is true about MQTT?

- Subscribes through topic patterns
- Use / as a separator in topics
- The + wildcard matches a single level
- The # wildcard matches all levels
- Supports QoS 0, 1 and 2
- Support last will and testament
- Support persistent sessions

To get messages for the following topics in MQTT, what is the correct subscription for retrieving all temperature measurements?

Topics:

- "home/temperature/sensor1",
 - "home/temperature/sensor2",
 - "home/humidity/sensor1",
 - "home/humidity/sensor2",
 - "home/pressure"
-
- "home/temperature/#"
 - "home/temperature/+"

To get messages for the following topics in MQTT, what is the correct subscription for retrieving all measurements from sensor1?

Topics:

- "home/temperature/sensor1",
 - "home/temperature/sensor2",
 - "home/humidity/sensor1",
 - "home/humidity/sensor2",
 - "home/pressure"
-
- "home/+ /sensor1"
 - "home/+ /sensor1/#"

What is correct about system failures?

- As the number of machines in a system goes towards infinity the time to one of them breaking goes to zero
- You have to deal with failures
- Built redundancy across systems and machines
- BEAM can be used for homogeneous and high-granularity systems
- Kubernetes, Ceph, Docker swarm can be used for heterogeneous and low-granularity systems

What is a system?

- A collection of independent components that has been combined in a particular configuration in order to solve some problem

What is true about hypothesis and hypothesis testing?

- Expectations can be formulated as a set of hypothesis
- Hypothesis testing is the act of testing to which degree a hypothesis is true
- Results of hypothesis testing is usually boolean, floats or probabilistic
- Hypothesis testing has its roots in statistics
- Hypothesis testing may be done on a subset of a system

What is true about tests in hypothesis testing?

- They are designed to highlight relevant performance metrics of the phenomenon of interest
- They are administered using experimental setup surrounding the apparatus

- They can be done manually or automatically
- They need to be repeatable

What is a full-factorial experiment?

- Examining all combinations of factors
- Use all possible parameters for the experiment
- An computationally expensive type of test

What is some elements an experiment should

- Include the stubs necessary to execute the system as defined
- Operate instances of the relevant components according to fixed values for all parameters
- Include stub implementation for all dependencies to the system
- Produce the needed workload
- Iterate over relevant factors and workloads
- Instrument relevant components to observe and log relevant metrics
- Not influence the system under experimentation

When testing, what is true about time?

- High-resolution timing is expensive
- High-resolution timing is often not critical
- Timestamps across machines are often not directly comparable
- Timestamps across a single machine may not be directly comparable
- Monotonic clocks can be helpful for short periods of time
- It is important to understand and document how large the time difference is between multiple systems

What are some key things about presenting plots?

- Any presented plot of data should be self-contained
- Context or context reference should be obvious
- Any axis in the plot should be labeled with metric and unit
- Ensure axis starts at appropriate zero or known value

In IoT, what is the definition of "budget"

- Represents the availability of a scarce resource
- Can be consumed
- Some get replenished
- Can be a defining factor for operation of a device

What are some restrictions in regards to radio budgets?

- Number of packets sent throughout a day
- Rate of transmission
- Maximum packet size

What are some restrictions in regards to memory budgets?

- Available size of RAM
- Available size of ROM
- Available size of flash
- Number of per-cell write cycles
- Size of erase unit

What is true about budget management?

- There are lots of budgets
- There are lots of tradeoffs
- It is hard to reason about the relative/absolute magnitudes of tradeoffs
- It is hard to reason about how design choices affect budgets
- There is no one-size-fits-all solution for budget management
- You should experiment with different budgets, to identify the best tradeoffs for your system

What is true about data throttling?

- If one or more values are not sent, the data may not be able to be reconstructed
- Blowing the budget, may cause data to not be sent

Which of the following is correct about sending aggregated values?

- You will get a sum curve that grows indefinitely
- You will transmit the same number of values
- You will be able to easily derive the original values from the aggregated values

What happens when you send aggregated values, and loose a value?

- Data degrades gracefully in temporal resolution

What is adaptive sampling?

- The frequency of sampling is adjusted based on the current state of the system
- The frequency is adjusted when changes in the system are detected

When working with data throttling, what is true?

- Batch transmission can be used to limit number of empty packets
- Batch transmission can limit power usage
- Batch transmission can increase latency of data
- It takes time going from sleep to active state

In order to reduce power when operating radio communication, what can be done?

- Low power listening
- Receiver listens in short intervals for transmissions
- Transmitter waits for receiver to be ready before transmitting

Of the following, what is correct?

- Naming has everything to do with being able to reference something
- Identity is the same as a name
- A device may have multiple identities

What are some sources of identity?

- Unique device number
- User provided
- Physical location
- Network-derived
- Context-derived

What are possible places to store identity?

- Node - By the device itself (self-aware)
- Cloud - Central knowledge base
- Private - Stored with the user

When are possible points to get a timestamp?

- When sampling on device
- When transmitting
- When receiving
- When pushing to cloud
- When receiving in the cloud
- When receiving from the cloud

What is Spread spectrum and frequency hopping?

- Originally invented to communicate between controllers and torpedoes more robust
- Signal spread over multiple spectrums
- Used in satellite communication
- Enhances robustness against interference

What are some of the pros and cons of using Spread spectrum and frequency hopping?

- Security increased
- Less susceptible to interference
- More robust against jamming
- Limits detection of communication
- Short range

What is beamforming?

- Antenna array that can focus the signal in a specific direction
- Focussing done by controlling signal phases and amplitude across multiple antennas

- Utilizes constructive and destructive interference

What are some of the pros and cons of using beamforming

- Improved signal strength
- Energie efficient
- Reduced interference
- Greater range
- Faster data rates
- Cost
- Complexity

What is backscattering, and what is it used for?

- A technique to communicate by reflecting existing signals
- Reflected signal is modulated in amplitude
- Used in RFID

What is SDR?

- Software-Defined Radio
- Allows for programmable hardware
- Allows for operating on multiple frequencies and protocols

Where can low power listening be found?

- Zigbee
- Bluetooth Low Energy (BLE)

When is mesh network topology usefull?

- Remote and difficult environments
- Disaster recovery and emergency situations
- Large events and festivals
- Military operations
- Smart cities and IoT applications

What is correct about eventual consistency?

- It's okay that different nodes have different versions of the same data
- All nodes will eventually converge to correct version of the data
- Quries might return old data
- PProblem: Maintaining data consistency across nodes